

A Framework for Estimating Road-Blockage Probability from Conflict Events

A supervised, parametric impact model

Conflict-event → Road-Blockage (C2RB) model

ACLED, Afghanistan, 2017–2026 (69,655 geocoded events)

1 Problem and approach

Goal. Given a conflict event (or a set of events over a time window), estimate which nearby roads are *affected* and the *probability that each road is blocked*. The central modelling question is: **how large is an event’s impact “buffer zone”, and how does it differ by sub_event_type?**

Ground-truth labels. Each event carries two human/LLM-verified yes/no labels: `is_road_affected` (the event impacted a road in any way — 20.8% of events) and `is_road_blocked` (a road was actually closed / made impassable — 0.87% of events); the latter is our calibration and scoring target. (*The framework was originally built weakly-supervised — the signal was mined from the notes free text — because the raw export had no such column; that proxy has now been replaced by these labels.*)

Our approach — a supervised parametric probability model. We build an explicit, interpretable probability model whose parameters are (a) **seeded from published literature** (radii and decay shapes) and (b) **calibrated against the ground-truth `is_road_blocked` label**. This is defensible, reproducible, and mirrors established practice: ACLED’s own *Conflict Exposure* product uses fixed event-type buffers (1/2/5 km) with Voronoi de-duplication, and a published conflict-impact model assigns per-event-type radii (Battles 25 km, Explosions 10 km, Protests/Riots 5 km) with type-specific decay shapes.

Pipeline. event $\longrightarrow R_{\text{eff}} \longrightarrow K(d) \longrightarrow P_{\text{block}}(d) \xrightarrow{\text{noisy-OR + time decay}} P(\text{road blocked})$

2 The model

For an event e with sub-event-type s , geo-precision code $g \in \{1, 2, 3\}$, fatalities n_f , and civilian-targeting flag $c \in \{0, 1\}$:

2.1 Effective buffer radius — “how big is the zone?”

$$R_{\text{eff}}(e) = \sqrt{R_{\text{phys}}(s)^2 + R_{\text{geo}}(g)^2} \cdot M_{\text{sev}}(e) \quad (1)$$

- $R_{\text{phys}}(s)$ — physical/operational impact radius of the event type (km), literature-seeded per `sub_event_type`. E.g. IED 1 km, armed clash 15 km, protest 4 km.
- $R_{\text{geo}}(g)$ — *locational uncertainty* of where the event actually happened. ACLED geo-precision 1 $\approx$ town (1 km), 2 $\approx$ near a town / part of a region (5 km), 3 $\approx$ wider region (25 km). A low-precision event must spread its blockage probability over a larger uncertain area.
- **Quadrature combination** $\sqrt{R_{\text{phys}}^2 + R_{\text{geo}}^2}$. Both terms behave like independent Gaussian length-scales (physical spread and locational error); variances of independent Gaussians add, so their standard deviations add in quadrature. This is more principled than naive addition (which double-counts) and never smaller than either term.

The severity multiplier $M_{\text{sev}}(e)$ stretches the radius for deadlier and civilian-directed events, with diminishing returns so a 50-fatality battle does not produce a $50\times$ radius:

$$S(n_f) = \alpha(1 + n_f) + (1 - \alpha)(1 + \ln(1 + n_f)) \quad (2)$$

$$M_{\text{sev}}(e) = \text{clip}\left(\left(1 + \kappa\left(\frac{S(n_f)}{S(n_{\text{ref}})} - 1\right)\right) \cdot (1 + \gamma_{\text{civ}} c), 0.5, 3.0\right) \quad (3)$$

with defaults $\alpha = 0.15$ (mostly log growth), $\kappa = 0.25$, $\gamma_{\text{civ}} = 0.30$. The `clip` bounds keep outliers from exploding the buffer.

Explanation of Equation (2) (the severity score $S(n_f)$). A deadlier event tends to affect a larger area, but not linearly — the jump from 0 to 5 fatalities matters far more than 100 to 105. Equation (2) captures this by **blending two growth curves**:

- the **linear term** $(1 + n_f)$ grows in direct proportion to the fatality count — it lets genuinely large, mass-casualty events widen the buffer; and
- the **logarithmic term** $(1 + \ln(1 + n_f))$ grows ever more slowly as n_f rises — it applies *diminishing returns* so extreme counts do not blow up the radius without bound.

The mixing weight $\alpha \in [0, 1]$ sets the balance: $\alpha = 0$ is pure-log (gentle), $\alpha = 1$ is pure-linear (aggressive); the default 0.15 keeps growth mostly logarithmic with a small linear component. The $1+$ offsets are deliberate — they make $\ln(1 + n_f)$ well-defined at $n_f = 0$ (avoiding $\ln 0 = -\infty$) and guarantee $S(0) = 1$, so a zero-fatality event is the natural baseline. S enters Equation (3) only through the *ratio* $S(n_f)/S(n_{\text{ref}})$: the event is compared to a reference fatality level n_{ref} (default 1), the ratio is turned into a gentle multiplier by κ , bumped by γ_{civ} if civilians were targeted, and the product is bounded to $[0.5, 3]$. In short, Equation (2) converts a raw body count into a *bounded, diminishing-returns severity signal* that stretches the buffer sensibly rather than mechanically.

2.2 Distance-decay blockage probability — “how does it fall with distance?”

$$P_{\text{block}}(d, e) = P_0(s) \cdot K_s(d; R_{\text{eff}}) \quad (4)$$

- $P_0(s)$ — *peak blockage propensity*: the probability that an event of type s blocks a road at the epicentre. **This is the parameter calibrated on the `is_road_blocked` label** (Sec. 3).
- $K_s(d) \in [0, 1]$ — distance-decay kernel, its *shape* chosen by the event type’s physics (Table 1).

2.3 Combining many events — noisy-OR with temporal decay

A road segment near several events is blocked if *any* of them blocks it. Treating events as independent blocking attempts gives the **noisy-OR**, with an **exponential temporal decay** $\varphi(\Delta t) = 0.5^{\Delta t/\tau}$ (half-life $\tau = 30$ days) so older events fade:

$$P_{\text{segment}} = 1 - \prod_{e \in W} \left[1 - P_{\text{block}}(d_e, e) \cdot \varphi(\Delta t_e) \right] \quad (5)$$

over a time window W . ACLED-style Voronoi de-duplication can be layered on to avoid double-counting tightly-clustered events (noted as an extension).

2.4 From points to roads

We use the WFP/AGCHO Afghanistan road network (`afg_trs_roads_1_arazi_2023x_fixed_segmented.shp`): **27,991 LineString segments, ≈ 1 km each** (median 1,068 m), EPSG:4326, each carrying road name, class, average slope and altitude. Each segment is represented by its midpoint; d_e is the segment-to-event distance. Every segment within $r_{\text{max}} = 3 R_{\text{eff}}$ of an event receives that event’s $P_{\text{block}}(d_e, e)$, and segments are aggregated by noisy-OR. The result is a per-segment blockage probability that rolls up to **named roads**. Without a road layer the same scoring runs over a

Kernel	Formula	Used for	Rationale
Gaussian	$\exp\left(-\frac{d^2}{2\sigma^2}\right)$, $\sigma = R_{\text{eff}}/2$	IED, suicide, grenade, shelling, airstrike	sharp, blast-like, fast decay
Exponential	$\exp\left(-\frac{d}{\lambda}\right)$, $\lambda = R_{\text{eff}}/3$	armed clash / battles	heavier tail — fighting spreads along ter- rain/roads
Uniform	$\mathbf{1}\{d \leq R_{\text{eff}}\}$	protests, riots, territorial control	roughly constant over an area, then cut off
Point	$\mathbf{1}\{d \leq \max(R_{\text{eff}}, 0.3)\}$	targeted violence, arrests	effect con- fined to the site

Table 1: Distance-decay kernels, chosen per event type.

regular lat/lon grid, producing a continuous blockage-risk raster (the fallback) — the code path is identical, only the target points differ.

3 Calibration & validation on the ground-truth labels

3.1 The two labels — and why we model *blocked*, not *affected*

Each event carries two yes/no labels: `is_road_affected` (14,456 events, 20.8%) and `is_road_blocked` (607 events, 0.87%). Blockage is almost a strict subset of affectedness — only **1 of 607** blocked events is not also flagged affected — so the labels are internally consistent ($P_0^{\text{blk}} \leq P_0^{\text{aff}}$ by construction) and $P(\text{blocked} | \text{affected}) = 0.042$. The two tell *different stories*, which is exactly why we calibrate on *blocked*:

sub_event_type	n	P(affected)	P(blocked)
Remote explosive/landmine/IED	8,545	0.656	0.003
Suicide bomb	289	0.405	0.000
Violent demonstration	44	0.386	0.182
Peaceful protest	1,454	0.102	0.061
Air/drone strike	6,249	0.049	0.001

Table 2: Affected $\neq$ blocked: an IED *affects* roads constantly (roadside bomb) but rarely *blocks* them; a demonstration is the reverse.

An IED/landmine *affects* roads constantly (66% of the time) but rarely *blocks* them (traffic resumes — 0.3%); a protest or demonstration deliberately *blocks* the road. A single “is the road relevant?” signal conflates these; the ground-truth `is_road_blocked` label separates them cleanly.

3.2 Peak blockage propensity $P_0(s)$

$P_0(s) = P(\text{is_road_blocked} = \text{yes} \mid s)$ is estimated with **Beta–Binomial empirical-Bayes shrinkage** (prior strength 50, anchored at the 0.87% global rate) so small-sample types (Grenade $n=176$ with a single blocked event, Suicide bomb $n=289$ with none) get stable estimates instead of noisy near-zero rates:

$$\hat{P}_0(s) = \frac{k_s + a_0}{n_s + a_0 + b_0}, \quad a_0 = \bar{p}m, \quad b_0 = (1 - \bar{p})m, \quad m = 50, \quad (6)$$

where k_s positives out of n_s events of type s , and $\bar{p}$ is the global rate. Selected results with Beta posterior 95% credible intervals (full 24-row table with credible intervals in `artifacts/afghanistan/model_card.m`) the P_0 values themselves are written to `countries/afghanistan.yaml`):

sub_event_type	n	P_0 (shrunk)	95% CrI
Violent demonstration	44	0.090	[0.04, 0.15]
Change to group/activity	318	0.083	[0.06, 0.11]
Non-violent transfer of territory	130	0.080	[0.05, 0.12]
Peaceful protest	1,454	0.060	[0.05, 0.07]
Non-state actor overtakes territory	905	0.060	[0.05, 0.08]
Armed clash	39,974	0.007	[0.006, 0.008]
Remote explosive/landmine/IED	8,545	0.003	[0.002, 0.005]
Air/drone strike	6,249	0.001	[0.000, 0.002]

Table 3: Calibrated blockage propensity — the headline result.

Headline result. Road *blockage* is driven by **protests and territorial-control events** (deliberate, sustained closures), not by explosive violence — the *opposite* of an affected-based ranking. This confirms that the model **must** vary by `sub_event_type`, and shows why the ground-truth label matters: it corrects the old weak label, which wrongly placed IEDs at the top by conflating *affected* with *blocked*.

3.3 Validation — supervised, on the true label

A logistic regression $\text{is_road_blocked} \sim \text{sub_event_type} + \log(1 + \text{fatalities}) + \text{geo_precision} + \text{civilian_targeting}$, evaluated with 5-fold cross-validation:

- **AUC** = 0.73 — event attributes discriminate road-blocking events well above chance.
- **PR-AUC** = 0.042 vs a 0.0087 prevalence — $\sim 5\times$ better than random under heavy imbalance.
- **Brier** = 0.0085 — *better* than the no-skill baseline (0.0086), i.e. genuinely informative.
- **Well-calibrated** — predicted $\approx$ observed across deciles. Use the plain model for calibrated probabilities; `class_weight="balanced"` improves ranking but inflates probabilities (apply isotonic/Platt calibration if class-weighting).

The model is both discriminative and calibrated, so the $P_0(s)$ values are trustworthy as probabilities, not merely rankings.

3.4 Other sanity checks (in the notebook)

- **Roads-within-1 km benchmark — passed.** With the real road network loaded, **78% of events fall within 1 km of a road**, independently matching the published $\sim 70\%$ figure — strong external validation.

- **Nesting check** — blocked $\subseteq$ affected (1 exception in 607), confirming label consistency.
- **Sensitivity analysis** over R_{phys} (all physical radii scaled 0.5–2 $\times$) shows how the high-risk footprint (grid cells with $P > 0.3$) responds to assumptions (what is assumed vs. learned). Kernel shapes are not swept in the notebook.

4 Statistical methods used

Distance-decay kernels (Gaussian / exponential / uniform); quadrature combination of independent uncertainty scales; bounded log-linear severity scaling; noisy-OR probabilistic aggregation with exponential temporal decay; Beta–Binomial empirical-Bayes shrinkage with Beta posterior credible intervals (and Wilson intervals on raw rates); supervised logistic regression with cross-validated AUC / PR-AUC / Brier / calibration-curve diagnostics under class imbalance; (optional) Voronoi tessellation for overlap de-duplication and KDE for hotspot context.

5 Limitations & honest caveats

1. **Labels are ground truth, but derived from notes.** `is_road_affected` / `is_road_blocked` were classified from the ACLED `notes` text, so they inherit its reporting quality — an event whose notes omit a closure can be a false negative. **Class imbalance is severe** (blocked = 0.87%), so absolute probabilities are small and PR-style metrics matter more than accuracy. Cross-checking against independent closure data (OCHA/Logistics Cluster access reports) would further harden P_0 — the calibration code is unchanged.
2. R_{geo} **cannot be learned from this data.** ACLED snaps all events at a named location to one shared centroid, so intra-location coordinate spread is structurally ~ 0 km. We therefore fall back to ACLED’s *documented* precision semantics (1/5/25 km). With finer coordinates, the empirical-spread diagnostic in the notebook (§4) would show a non-zero spread, which can then be set as `r_geo`.
3. R_{phys} **and kernel shapes are assumptions** (literature-seeded), unlike P_0 (data-calibrated). The sensitivity analysis quantifies how much R_{phys} matters (kernel shapes are not yet swept); tune in `params.base.yaml`.
4. **Independence in noisy-OR** slightly over-counts spatially correlated events; Voronoi de-duplication (ACLED’s approach) is the recommended refinement.
5. **Static / impact-only.** This scores the impact of given events; it does not forecast where future events will occur.

6 Pipeline, configuration, and localization

The framework runs headless through three CLI scripts, all reading a *layered* configuration. `python train.py --config countries/<name>.yaml` calibrates $P_0(s)$ on the country’s `is_road_blocked` labels (Beta–Binomial shrinkage), writes it back into the country config, and emits `artifacts/<name>/` (`model.joblib`, `metrics.json`, `model_card.md`, calibration plots). Training produces *two models for two questions*: the calibrated parametric P_0 (the interpretable core that drives the spatial scorer), and a logistic-regression classifier that is cross-validated for the §3.3 metrics and then refit on all rows and persisted into `model.joblib` as the deployable event-level model. `python score.py --config countries/<name>.yaml --as-of <date> --window-days <n>` scores road segments via noisy-OR over the window’s events and ranks the most likely-blocked named roads, falling back to a grid surface if no road layer is set — the *spatial* question (*which roads* are blocked). For the *event-level* question (*will this event* block a road?), `python predict.py --config countries/<name>.yaml --sub-event-type ...` loads the deployable `model.joblib` — the classifier

`train.py` fits on the labels and persists alongside P_0 — and returns one probability per event (single event or CSV batch).

Country-independent physics — R_{phys} , decay kernels, R_{geo} , the severity and temporal constants — lives in `params.base.yaml` and is shared by every country. Each country adds a small override (`countries/<name>.yaml`) carrying only its data paths and calibrated P_0 (plus an optional `crs_metric` hint for GIS preprocessing); `c2rb.load_config` merges the two. Adapting the model to a new country means copying the override, pointing it at that country’s ACLED export and segmented road network, and re-running `train.py` then `score.py` — no changes to the `c2rb` package. See `docs/COUNTRY_GUIDE.md` for the procedure (including `tools/segment_roads.py`, which cuts a raw road network into ~ 1 km segments).

File	Purpose
<code>c2rb/</code>	Framework core package: geometry, calibration, config, model physics, spatial scorer, classifier
<code>train.py</code> / <code>score.py</code> / <code>predict.py</code>	CLI: calibrate + save the model / per-road rankings (spatial) / event-level $P(\text{blocked})$
<code>app.py</code>	Streamlit app: event-level prediction + click-to-place spatial scoring
<code>artifacts/<name>/model.joblib</code>	Deployable trained model: event-level classifier + parametric P_0
<code>params.base.yaml</code>	Universal parameters: R_{phys} , decay class, R_{geo} , severity & time constants (shared)
<code>countries/<name>.yaml</code>	Per-country config: data paths, calibrated P_0 (written by <code>train.py</code>)
<code>docs/</code>	Stage-by-stage pipeline documentation (data → labeling → training → evaluation → deployment)
<code>docs/COUNTRY_GUIDE.md</code>	Step-by-step guide to building a model for another country
<code>tests/</code> / <code>Makefile</code>	Pytest smoke suite over the core math and config; make targets for the pipeline
<code>tools/segment_roads.py</code>	Utility: cut a raw road network into ~ 1 km segments
<code>conflict_road_blockage.ipynb</code>	Annotated end-to-end walkthrough: load → label → calibrate → score → map → sensitivity
<code>build_notebook.py</code>	Regenerates the notebook from cell definitions (diffable source of truth)
<code>requirements.txt</code>	Pinned dependencies for reproducibility
<code>methodology.tex</code> / <code>.pdf</code>	This document

Reproduce. Create the venv and `pip install -r requirements.txt`, then either run the CLI scripts above or execute the notebook top-to-bottom. The road and admin layers are already wired in `countries/afghanistan.yaml`; set `paths.roads: null` to fall back to grid-raster mode.

7 References

- ACLED, *Conflict Exposure methodology* (buffer zones 1/2/5 km, event-type adjustment, Voronoi de-duplication).
<https://acleddata.com/methodology/conflict-exposure-methodology>
- ACLED, *Codebook* (event types, sub-event types, geo-precision codes) — `AcledCodebook.md`.
- Applying Computational Engineering Modelling to Analyse the Social Impact of Conflict and*

Violent Events (per-type impact radii & decay shapes), PMC12562815.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12562815/>

- *Proximity to roads shapes patterns of violence in North and West Africa* (~70% of violent events within 1 km of a road).
<https://anl.geog.ufl.edu/roads-violence-nwafrica/>
- Road network: WFP / AGCHO-NSIA Afghanistan segmented roads (`afg_trs_roads_1_arazi_2023x`); admin boundaries: AGCHO-NSIA Afghanistan adm1/adm2 (2017).